

# Heisenberg Group and Reality

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## 1

Let  $H$  be the complex Heisenberg group, an extension of  $\mathbb{C}^2$  by  $\mathbb{C}^*$ . It has coordinates  $(x, y, \lambda)$ , the group law being

$$(x, y, \lambda)(x', y', \lambda') = (x + x', y + y', \lambda\lambda' \exp(\pi i(xy' - x'y))). \quad (1.1)$$

The involution

$$\tau : (x, y, \lambda) \mapsto (-x, -y, \lambda) \quad (1.2)$$

is an automorphism of  $H$ .

The coordinate system  $x, y, \lambda : H \rightarrow \mathbb{C}^2 \times \mathbb{C}^*$  exhibits the section  $\mathbb{C}^2 \times \{1\}$  of the projection of  $H$  onto  $\mathbb{C}^2$ . On this section,  $\tau(g) = g^{-1}$ . Each line  $D \subset \mathbb{C}^2$  has, by means of this section, a lift to a one-parameter subgroup of  $H$ , again denoted by  $D$ . The commutator  $(u, v) = uvu^{-1}v^{-1}$  of two elements  $u = (u_1, u_2)$  and  $v = (v_1, v_2)$  of  $\mathbb{C}^2$  is  $\exp(2\pi i \langle u, v \rangle) \in \mathbb{C}^*$ , with

$$\langle u, v \rangle = u_1 v_2 - u_2 v_1. \quad (1.3)$$

Let  $H_{\mathbb{R}}$  be the real form defined by  $x, y \in \mathbb{R}$  and  $|\lambda| = 1$ . The group  $\mathrm{SL}(2, \mathbb{C})$  acts on  $H$ , and  $\mathrm{SL}(2, \mathbb{R})$  acts on  $H_{\mathbb{R}}$ , which as a set is identified with  $\mathbb{R}^2 \times U^1$ .

It is known, by the Stone-von Neumann theorem [2], that  $H_{\mathbb{R}}$  has a unique irreducible unitary representation  $V$  for which  $\lambda \in U^1$  acts by multiplication by  $\lambda$ . In the Schrödinger model of  $V$ , one has  $V = L^2(\mathbb{R})$ , and the action of  $H_{\mathbb{R}}$  is

$$\begin{aligned} (x, 0, 1) : f(z) &\mapsto f(z + x), \\ (0, y, 1) : f(z) &\mapsto e^{2\pi i y z} f(z). \end{aligned} \quad (1.4)$$

From the uniqueness of  $V$  one deduces a projective unitary action of  $\mathrm{SL}(2, \mathbb{R})$  on  $V$ , compatible with the action of  $\mathrm{SL}(2, \mathbb{R})$  on  $H_{\mathbb{R}}$  (A. Weil [3], D. Shale [1]).

For the action of  $H_{\mathbb{R}}$  on  $V = L^2(\mathbb{R})$ , the  $C^\infty$  vectors, i.e. those  $v \in V$  for which  $h \mapsto hv$  is  $C^\infty$  in  $h \in H_{\mathbb{R}}$ , form the Schwartz space  $\mathcal{S}$  of  $C^\infty$  rapidly decreasing functions on  $\mathbb{R}$ . The holomorphic vectors, i.e. those  $v \in V$  for which  $h \mapsto hv$  extends to a holomorphic map from  $H$  to  $V$ , are the  $L^2$  functions on  $\mathbb{R}$  that extend to holomorphic functions on  $\mathbb{C}$  such that, in every horizontal strip  $|\mathrm{Im} z| < A_1$ , one has, for all  $A_2 \in \mathbb{R}$ ,

$$|f(z)| \leq O(\exp(-A_2|z|)).$$

The complex group  $H$  acts on the space  $V^{\mathrm{hol}}$  of holomorphic vectors by the formulae (1.4).

For the holomorphic action of  $H$  on  $V^{\mathrm{hol}}$ , one does not have the same uniqueness as for the unitary action of  $H_{\mathbb{R}}$  on  $V$ . If one did, the projective action of  $\mathrm{SL}(2, \mathbb{R})$  on  $V$  would extend to a projective action of  $\mathrm{SL}(2, \mathbb{C})$  on  $V^{\mathrm{hol}}$ ; since  $\mathrm{SL}(2, \mathbb{C})$  is simply connected, this action would lift to an honest action of  $\mathrm{SL}(2, \mathbb{C})$ , and the projective action of  $\mathrm{SL}(2, \mathbb{R})$  would lift to an honest action. This is not the case: only the double cover of  $\mathrm{SL}(2, \mathbb{R})$  acts. Our purpose is to make explicit what happens.

## 2

Let  $S = \mathbb{P}^1(\mathbb{C})$  be the space of lines of  $\mathbb{C}^2$ . For  $D \in S$ , lifted in  $H$  as in no. 1, let  $W_D$  be the space of holomorphic functions  $\varphi$  on  $H$  satisfying

$$\varphi(\lambda dh) = \lambda \varphi(h)$$

for  $\lambda \in \mathbb{C}^*$  and  $d \in D$ , endowed with the action of  $H$  given by  $g : \varphi(h) \mapsto \varphi(hg)$ . For  $E \neq D$ , restriction to  $E$  is a bijection

$$r_E : W_D \xrightarrow{\sim} \mathcal{O}(E), \quad (2.1)$$

where  $\mathcal{O}(E)$  denotes the space of holomorphic functions on  $E$ .

If  $R$  is a holomorphic representation of  $H$ , on which each  $\lambda \in \mathbb{C}^*$  acts by  $\lambda$ , and if  $\omega$  is a  $D$ -invariant linear form on  $R$ , then

$$w \mapsto \omega(hw) \quad (2.2)$$

is a morphism from  $R$  to  $W_D$ .

## 3. Example

Let  $V$ , as in nos. 1 and 2, be the irreducible unitary representation of  $H_{\mathbb{R}}$ , let  $\mathcal{S} \subset V$  be the space of its  $C^\infty$  vectors, and let  $\mathcal{S}' \supset V$  be the dual of the space of  $C^\infty$  vectors of the dual representation; in the Schrödinger model, this is the space of tempered distributions.

Let  $D_t \in S$  be the line generated by  $(-t, 1)$ . If  $\text{Im } t > 0$ , the linear form, written in the Schrödinger model,

$$\omega_t : f \mapsto \frac{1}{\sqrt{t/i}} \int_{\mathbb{R}} f(x) \exp(-\pi i x^2/t) dx$$

is defined on  $V$ , and even on  $\mathcal{S}'$ . The form  $\omega_t$  is a holomorphic vector of the dual of  $V$ , fixed by  $D_t$ . In terms of the infinitesimal generator  $-t\partial_x + 2\pi i x$  of  $D_t$ , the invariance is written

$$\int_{\mathbb{R}} [(-t\partial_x + 2\pi i x)f(x)] \exp(-\pi i x^2/t) dx = \int_{\mathbb{R}} f(x)(t\partial_x + 2\pi i x) \exp(-\pi i x^2/t) dx = 0.$$

By (2.2),  $\omega_t$  defines

$$\Omega_t : V^{\text{hol}} \longrightarrow W_{D_t},$$

which extends to  $V$  and even to  $\mathcal{S}'$ . Let  $E$  be the  $x$ -axis. The map  $r_E \Omega_t : V \rightarrow \mathcal{O}(E)$  sends  $f \in V$  to

$$f(x, t) = r_E \Omega_t(f) \quad (\text{Im } t > 0),$$

a solution of the heat equation

$$\left( \partial_t - \frac{1}{4\pi i} \partial_x^2 \right) f(x, t) = 0.$$

For real  $t$ , the form  $\omega_t$  remains defined on  $\mathcal{S} \subset V$ . For  $t = 0$  it is  $f \mapsto f(0)$ , and for  $t = \infty$  it is  $f \mapsto \int f(x) dx$ . The homomorphism  $\Omega_t$  remains defined on  $V^{\text{hol}}$ . For  $t = 0$ ,  $r_E \Omega_0$  is, in the Schrödinger model, the identity.

Assume  $\text{Im } t > 0$ . Let  $\overline{D}_t$  be the complex conjugate of  $D_t$ . In the affine line  $S - \{\overline{D}_t\}$ , the “half-plane”  $\{D_u \mid \text{Im } u > 0\}$  is a disk with center  $D_t$ . The map  $r_{\overline{D}_t} \circ \Omega_t : V \rightarrow W_{D_t} \rightarrow \mathcal{O}(\overline{D}_t)$  gives the holomorphic model of  $V$ : the action of  $H$  is given by

$$\begin{aligned} d \in D_t : \quad & f(x) \mapsto f(x + d), \\ e \in \overline{D}_t : \quad & f(x) \mapsto \exp(-2\pi i \langle e, x \rangle) f(x), \end{aligned}$$

and the Hilbert structure for which  $(f, f)$  is the integral of

$$f(d)\overline{f(\bar{d})}\exp(2\pi i\langle d, \bar{d} \rangle)$$

is invariant under  $H_{\mathbb{R}}$ ; moreover  $r_{\bar{D}_t} \circ \Omega_t$  identifies  $V$  with the space of  $f$  of finite norm. For  $d = (-tz, z)$ , the factor  $\exp(2\pi i\langle d, \bar{d} \rangle)$ , the commutator of  $d$  and  $\bar{d}$ , is

$$\exp(2\pi i\langle d, \bar{d} \rangle) = \exp(2\pi i(t - \bar{t})|z|^2).$$

## 4

Fix  $E$ . The isomorphisms  $r_E$  of (2.1) make it possible to regard the  $W_D$  for  $D \neq E$  as a family of representations of  $H$  operating in the fixed space  $\mathcal{O}(E)$ , with the action of  $H$  depending holomorphically on  $D$ . Let us make explicit the dependence on  $E$  of  $r_E$ . Choose coordinates as in no. 1, with  $E$  the  $x$ -axis. Let  $E'$  be generated by  $(1, u)$  and  $D$  by  $(-t, 1)$ . We use  $x$  as coordinate on both  $E$  and  $E'$ . Then

$$(x', ux', 1) = (-tux', ux', 1)(x, 0, 1)(0, 0, \lambda)$$

for  $(1 + tu)x' = x$  and  $\lambda = \exp(\pi i u x x')$ . Hence  $r_{E'} r_E^{-1}$  sends  $f_E \in \mathcal{O}(E)$  to  $f_{E'} \in \mathcal{O}(E')$ , with  $f_{E'}(x') = \lambda f_E(x)$ :

$$f_{E'}\left(\frac{x}{1+tu}\right) = \exp\left(\frac{\pi i u}{1+tu}x^2\right) f_E(x). \quad (4.1)$$

This formula is holomorphic in  $t$ : the  $W_D$  form a holomorphic vector bundle  $\mathcal{W}$  of infinite dimension over  $S$ , on which  $H$  acts.

To understand (4.1), one may observe that the point of  $E$  with coordinate  $x$  has the same image in  $\mathbb{C}^2/D$  as the point of  $E'$  with coordinate  $x/(1 + tu)$ .

From (4.1) one deduces that the following growth condition on  $v \in W_D$  is independent of the choices of  $E \neq D$  and of the coordinate  $z$  on  $E$ :

$$\text{for } f = r_E(v), \quad \exists A \quad |f(z)| \leq O(\exp(A|z|^2)). \quad (4.2)$$

Let  $W_D^0 \subset W_D$  be defined by this condition. The  $W_D^0$  form a holomorphic sub-bundle  $\mathcal{W}^0$  of the bundle  $\mathcal{W}$  in holomorphic representations of  $H$ .

## 5. Proposition

Let  $T > 0$  and consider the following conditions on an entire function  $f$ :

- (i)<sub>T</sub> There exists a holomorphic function  $g(z, t)$ , for  $z \in \mathbb{C}$  and  $|t| < T$ , satisfying the heat equation  $(\partial_t - \partial_z^2)g = 0$ , with initial condition  $g(z, 0) = f(z)$ .
- (ii)<sub>T</sub>  $|f(z)| \leq O(\exp(|z|^2/4T))$ .
- (iii)<sub>T</sub>  $\int f(z)\overline{f(z)}\exp(-|z|^2/2T)|dz \wedge d\bar{z}| < \infty$ .

If  $T < T_1$ , each of  $(i)_{T_1}$ ,  $(ii)_{T_1}$ ,  $(iii)_{T_1}$  implies each of  $(i)_T$ ,  $(ii)_T$ ,  $(iii)_T$ .

*Proof.*  $(i)_{T_1} \Rightarrow (ii)_T$ . The heat equation remains satisfied by the derivatives of  $g$ , and therefore

$$\partial_z^{2n} g = \partial_t^n g, \quad \partial_z^{2n+1} g = \partial_t^n (\partial_z g).$$

Since  $g(0, t)$  is holomorphic for  $|t| < T_1$ , if  $T < T_2 < T_1$ , then at  $(0, 0)$

$$|\partial_t^n g/n!| \leq O(1/T_2^n), \quad |\partial_z^{2n} g| \leq O(n!/T_2^n),$$

and there is an analogous estimate for  $\partial_z^{2n+1}g$ . The Taylor series of  $f$  then gives

$$|f(z)| \leq O\left(\sum \frac{n!}{(2n)!} \frac{1}{T_2^n} |z|^{2n}\right).$$

One has

$$\frac{1}{2n+1} 2^{2n} \leq \frac{(2n)!}{(n!)^2} \leq 2^{2n},$$

hence

$$|f(z)| \leq O\left(\sum 2^{-2n} \frac{1}{T^n} \frac{|z|^{2n}}{n!}\right),$$

and this sum is  $\exp(|z|^2/4T)$ .

$(ii)_{T_1} \Rightarrow (i)_T$ . It suffices to take  $g$  given by the kernel of the heat equation:

$$g(z, t) = \int_{\sqrt{t}\mathbb{R}} f(z+x) \frac{1}{2\sqrt{\pi t}} \exp(-x^2/4t) dx.$$

Changing  $\sqrt{t}$  to  $-\sqrt{t}$  changes the orientation of the cycle  $\sqrt{t}\mathbb{R}$  and the sign of the integrand, so that the integral does not change. Condition (ii) ensures convergence for  $|t| < T$ .

The relation between (ii) and (iii) is clear:  $(ii)_{T_1} \Rightarrow (iii)_T$ , and  $(iii)_{T_1} \Rightarrow (ii)_T$  is verified by writing  $f(z)$  as the mean of its values on a disk with center  $z$  and radius  $1/|z|$ , and estimating that mean by Cauchy–Schwarz.

## 6

In homage to the Stone-von Neumann theorem, the bundle  $\mathcal{W}$  over  $S$  is, projectively, equipped with a holomorphic connection for which the action of  $H$  is horizontal.

Let us specify “projectively.” On the projective line  $S$  there is the ample invertible sheaf  $\mathcal{O}(1)$ . Let  $\mathcal{O}(-1)$  be its dual. A square root  $\mathcal{O}(-\frac{1}{2})$  of  $\mathcal{O}(-1)$  is an invertible sheaf  $L$  endowed with an isomorphism  $L^{\otimes 2} \simeq \mathcal{O}(-1)$ . Locally such a square root exists and, locally, two square roots are isomorphic; but the isomorphism is not unique, with ambiguity  $\pm 1$ . This ambiguity obstructs the global existence of  $\mathcal{O}(-\frac{1}{2})$ . It is not on  $\mathcal{W}$ , but on

$$\mathcal{W}(-\tfrac{1}{2}) := \mathcal{W} \otimes \mathcal{O}(-\tfrac{1}{2})$$

that there is a connection:

$$\nabla : \mathcal{W}(-\tfrac{1}{2}) \longrightarrow \mathcal{W}(-\tfrac{1}{2}) \otimes \Omega^1. \quad (6.1)$$

Note that if  $\mathcal{O}(-\frac{1}{2})'$  and  $\mathcal{O}(-\frac{1}{2})''$  are two square roots of  $\mathcal{O}(-1)$ , and if  $\nabla$  is a connection on  $\mathcal{W} \otimes \mathcal{O}(-\frac{1}{2})'$ , then its transport to  $\mathcal{W} \otimes \mathcal{O}(-\frac{1}{2})''$  by an isomorphism  $a : \mathcal{O}(-\frac{1}{2})' \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})''$  of square roots of  $\mathcal{O}(-1)$  does not depend on the choice of  $a$ , which can be changed locally only by a constant  $\pm 1$ . Thus the notion of a “connection on  $\mathcal{W}(-\frac{1}{2})$ ” is well defined independently of the global existence, or the choice, of  $\mathcal{O}(-\frac{1}{2})$ .

Let  $D \in S$  and let  $t$  be a tangent vector at  $D$ . For a local section  $s$  of  $\mathcal{W}(-\frac{1}{2})$ , the problem is to define  $\nabla_t s$  in the fiber of  $\mathcal{W}(-\frac{1}{2})$  at  $D$ . We proceed as follows: after making an auxiliary choice  $u$ , one defines  $\nabla_t^u s$  for  $s$  a local section of  $\mathcal{W}$  or of  $\mathcal{O}(-\frac{1}{2})$ . This derivation  $\nabla_t^u$  depends on  $u$ , with dependence in  $u$  of the form

$$\nabla_t^{u'}(s) = \nabla_t^u(s) + as(D),$$

where  $a$  takes opposite values for  $\mathcal{W}$  and for  $\mathcal{O}(-\frac{1}{2})$ . Hence, for  $s = s_1 s_2$ , with  $s_1$  a local section of  $\mathcal{W}$  and  $s_2$  a local section of  $\mathcal{O}(-\frac{1}{2})$ ,

$$\nabla_t(s) := \nabla_t^u(s_1)s_2(D) + s_1(D)\nabla_t^u(s_2)$$

is independent of the choice of  $u$ .

Let  $D \in S$  be a line of  $\mathbb{C}^2$ . A tangent vector  $t$  at  $D \in S$  is identified with a linear map  $t : D \rightarrow \mathbb{C}^2/D$ , i.e. with an element of  $(\mathbb{C}^2/D)^{\otimes 2}$ : to  $u' \otimes v' \in (\mathbb{C}^2/D)^{\otimes 2}$  one associates  $t(d) = \langle u, d \rangle v'$ , where  $u$  is any representative of  $u'$ . The auxiliary datum will be a lift  $\tilde{t} : D \rightarrow \mathbb{C}^2$  of  $t$ , or equivalently a lift of  $t$  to  $\mathbb{C}^2/D \otimes \mathbb{C}^2$ . We write it  $\tilde{t} = u' \otimes u$ , with  $u \in \mathbb{C}^2$  mapping to  $u'$  in  $\mathbb{C}^2/D$ .

The tangent space at the origin of  $H$  is  $\mathfrak{h} = \mathbb{C}^2 \times \mathbb{C}$ . For  $z$  in this tangent space, let  $\delta_z$  be the corresponding right-invariant vector field. One has

$$[\delta_x, \delta_y] = -\delta_{[x,y]},$$

where on the right appears the bracket in the Lie algebra of  $H$ . For  $x, y \in \mathbb{C}^2$ , computing  $[x, y]$  as an infinitesimal commutator, one deduces

$$[\delta_x, \delta_y] = -2\pi i \langle x, y \rangle \delta_{(0,0,1)}. \quad (6.2)$$

Let  $\mathcal{U}_1$  be the quotient of the enveloping algebra of the Lie algebra of right-invariant vector fields by the relation  $\delta_{(0,0,1)} = 1$ . In  $\mathcal{U}_1$ , (6.2) implies that, for  $d \in D$ ,

$$\left[ -\frac{\delta_u^2}{4\pi i}, \delta_d \right] = \frac{1}{4\pi i} 2 \cdot 2\pi i \langle u, d \rangle \delta_u = \langle u, d \rangle \delta_u = \delta_{t(d)}. \quad (6.3)$$

Let us compute to first order around  $D$ . For  $\varepsilon^2 = 0$  and  $\varphi(g)$  in  $W_D$ , so  $\delta_d \varphi = 0$ , one has

$$\delta_{d+\varepsilon t(d)} \left[ \varphi(g) + \varepsilon \frac{-\delta_u^2}{4\pi i} \varphi(g) \right] = \varepsilon \left[ \delta_{t(d)} \varphi(g) + \delta_d \frac{-\delta_u^2}{4\pi i} \varphi(g) \right] = 0,$$

because  $\delta_d(-\delta_u^2/4\pi i)\varphi = -[-\delta_u^2/4\pi i, \delta_d]\varphi$ , by (6.3). To first order around  $D$ , the function of  $g$  and  $D'$  defined by

$$\varphi(g, D + \varepsilon t) = \varphi(g) + \varepsilon(-\delta_u^2/4\pi i)\varphi(g)$$

is therefore a section of the bundle  $\mathcal{W}$ . If  $\varphi(g, D')$  is a local section of  $\mathcal{W}$ , this permits one to define the derivative  $\nabla_t^u \varphi$  in  $\mathcal{W}_D$  by

$$\nabla_t^u \varphi = \delta_t \varphi(g, D') + (\delta_u^2/4\pi i)\varphi(g, D). \quad (6.5)$$

Let us compute the dependence of (6.5) on  $u$ . If  $u$  is replaced by  $u + x$  with  $x \in D$ , then in  $\mathcal{U}_1$

$$\delta_{u+x}^2 = (\delta_u + \delta_x)^2 = \delta_u^2 + 2\delta_u \delta_x + [\delta_x, \delta_u] = \delta_u^2 + 2\delta_u \delta_x + 2\pi i \langle u, x \rangle. \quad (6.6)$$

For  $\varphi \in W_D$ ,  $\delta_x \varphi = 0$ , and (6.6) becomes

$$\delta_{u+x}^2 \varphi = \delta_u^2 \varphi + 2\pi i \langle u, x \rangle \varphi, \quad (6.7)$$

whence

$$\frac{1}{4\pi i} \delta_{u+x}^2 \varphi = \frac{1}{4\pi i} \delta_u^2 \varphi + \frac{1}{2} \langle u, x \rangle \varphi. \quad (6.8)$$

The choice of  $u$  also makes it possible to define the derivative  $\nabla_t^u$  with respect to  $t$  of a local section of  $\mathcal{O}(1)$ . A section of  $\mathcal{O}(1)$  on  $U \subset S$  is a homogeneous function of degree 1 on the inverse image of  $U$  in  $\mathbb{C}^2$ , and one differentiates with respect to  $\langle u, d \rangle u$  to obtain a homogeneous function of degree 1 on  $D$ . When  $u$  is changed into  $u + x$ ,

$$\nabla_t^{u+x}(s) = \langle u + x, d \rangle \delta_{u+x} s = \nabla_t^u(s) + \langle u, x \rangle s.$$

This derivation induces one on  $\mathcal{O}(-\frac{1}{2})$ , with

$$\nabla_t^u(s^{-1/2}) = -\frac{1}{2} \nabla_t^u(s) s^{-3/2},$$

and, for the dependence on  $u$ ,  $s$  now being a local section of  $\mathcal{O}(-\frac{1}{2})$ ,

$$\nabla_t^{u+x}(s) = \nabla_t^u(s) - \frac{1}{2}\langle u, x \rangle s. \quad (6.9)$$

The dependences on  $u$  in (6.8) and (6.9) cancel to provide a derivation  $\nabla_t$  for  $\mathcal{W}(-\frac{1}{2})$ . Since the connection  $\nabla$  is defined in terms of right-invariant vector fields on  $H$ , it commutes with the action of  $H$ .

## 7

Let us spell out the connection of no. 6 in coordinates. Choose coordinates as in no. 1, and let  $E$  be the  $x$ -axis. On  $S$ ,  $\mathcal{O}(1)$  has a section  $s$  with a simple zero at  $E$ . It is unique up to a factor and trivializes  $\mathcal{O}(1)$  on  $S - E$ . Thus there is a square root  $\mathcal{O}(-\frac{1}{2})$  on  $S - E$ , trivialized by a square root  $s^{-1/2}$  of  $s^{-1}$ . By such a trivialization, the desired connection is transported to a connection on the bundle  $\mathcal{W}$ . We calculate it.

Let  $D_t$  be the line generated by  $(-t, 1)$ . By  $r_E$  of (2.1), a local section of  $\mathcal{W}$  is identified with a function  $f(z, t)$  holomorphic in  $z$  and  $t$ . We compute its derivative with respect to  $t$ . In no. 6, one may take

$$\tilde{t} : D_t \longrightarrow \mathbb{C}^2 : (-tz, z) \longmapsto (-1, 0)z; \quad \text{that is } d \longmapsto -\langle (1, 0), d \rangle (1, 0).$$

The corresponding derivation of  $\mathcal{O}(1)$  annihilates  $s$ , so that simply

$$\nabla_t f = (\partial_t - \partial_z^2 / 4\pi i) f, \quad (7.1)$$

and the horizontal local sections are the solutions of the heat equation

$$(\partial_t - \partial_z^2 / 4\pi i) f = 0. \quad (7.2)$$

From no. 5 one deduces that through  $s \in W_D^0(-\frac{1}{2})$  there passes a horizontal local section of  $\mathcal{W}(-\frac{1}{2})$ . It is automatically a section of  $\mathcal{W}^0(-\frac{1}{2})$ . It exists in a neighborhood of  $D$  that is larger the slower the growth of  $s$ . Note that in infinite dimension a connection does not necessarily define a local trivialization, even if through each point there passes a local horizontal section.

The connection  $\nabla$  on  $\mathcal{W}^0(-\frac{1}{2})$  defines a connection on the corresponding bundle of projective spaces

$$\mathbb{P}(\mathcal{W}^0) = \mathbb{P}(\mathcal{W}^0(-\frac{1}{2})).$$

The horizontal local sections are the images of the nonzero horizontal sections of  $\mathcal{W}^0(-\frac{1}{2})$ . Although the sphere  $S$  is simply connected, one has the following.

## 8. Surprise

The projective-space bundle  $\mathbb{P}(\mathcal{W}^0)$  on  $S$  has no global horizontal section.

The argument of no. 1 would have made it possible to predict this: apply it to the projective representation of  $\mathrm{SL}(2, \mathbb{C})$  given by the action of  $\mathrm{SL}(2, \mathbb{C})$  on the projective space of global horizontal sections of  $\mathbb{P}(\mathcal{W}^0)$ .

*First proof.* Divide  $S$  into two hemispheres  $S_1$  and  $S_2$ , and choose  $\mathcal{O}(-\frac{1}{2})_i$  on  $S_i$ . On  $S_1 \cap S_2$  one has locally an isomorphism  $\alpha : \mathcal{O}(-\frac{1}{2})_1 \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})_2$ , defined up to sign, but  $\alpha$  can be defined globally only on the double cover of the circle  $S_1 \cap S_2$ : the monodromy is  $-1$ . If there existed global horizontal projective sections of  $\mathbb{P}(\mathcal{W}^0)$ , there would exist horizontal sections  $f_1, f_2$  of  $\mathcal{W}^0(-\frac{1}{2})$  on  $S_1$  and  $S_2$  which, on  $S_1 \cap S_2$ , would coincide up to sign. On  $S_1 \cap S_2$

this would make it possible to normalize  $\alpha : \mathcal{O}(-\frac{1}{2})_1 \rightarrow \mathcal{O}(-\frac{1}{2})_2$  by  $\alpha(f_1) = f_2$ , contradicting the nonexistence of  $\alpha$  on  $S_1 \cap S_2$ .

*Second proof.* Choose coordinates as in no. 1. Let  $E$  be the  $x$ -axis and  $D$  the  $y$ -axis. As in no. 2, identify  $W_D$  with the space of entire functions  $f(z)$ . Let  $f \neq 0$  in  $W_D$ . By the proposition, for there to exist on  $S - E$  a horizontal section of  $\mathbb{P}(\mathcal{W}^0)$  passing through  $f$ , it is necessary and sufficient that  $f$  satisfy the growth condition

$$\forall \varepsilon > 0 \quad |f(z)| \leq O(\exp(\varepsilon|z|^2)). \quad (8.1)$$

If  $E'$  is another line, then by (4.1) the existence of a horizontal section on  $S - E'$  passing through  $f$  is expressed by a growth condition

$$\forall \varepsilon > 0 \quad |f(z) \exp(az^2)| \leq O(\exp(\varepsilon|z|^2)) \quad (8.2)$$

with suitable  $a \neq 0$ .

Put  $g(z) = f(z) \exp(az^2/2)$ . By (8.1) and (8.2),  $|g(z)|$  is bounded by a multiple both of  $|\exp(\varepsilon|z|^2 + az^2/2)|$  and of  $|\exp(\varepsilon|z|^2 - az^2/2)|$ ; hence in all directions except those for which  $az^2$  is purely imaginary,  $g$  decreases exponentially quadratically. In all directions, it has uniformly at most exponential-quadratic growth. By Phragmén–Lindelöf,  $g$  decreases exponentially quadratically in all directions, hence is zero, and therefore  $f = 0$ .

## 9

In the coordinates of no. 1, let  $D_t$  be the line generated by  $(-t, 1)$ . Let  $d = (-tz, z)$ . The commutator of  $d$  and  $\bar{d}$  is

$$\langle d, \bar{d} \rangle = \exp(-2\pi i(t - \bar{t})|z|^2) \in \mathbb{C}^*;$$

the commutator is real positive, and

$$\langle d, \bar{d} \rangle > 1 \iff \operatorname{Im}(t) > 0.$$

The lines  $D$  such that, for  $d$  generating  $D$ , one has  $\langle d, \bar{d} \rangle > 1$ , therefore form a disk in  $S$ , called the disk attached to the real structure  $H_{\mathbb{R}}$ .

The real structures  $L \subset \mathbb{C}^2$  for which  $\langle \cdot, \cdot \rangle$  is real on  $L$  form a homogeneous space under  $\operatorname{SL}(2, \mathbb{C})$ , with  $\operatorname{SL}(2, \mathbb{R})$  as the stabilizer of  $\mathbb{R}^2$ . Each defines a real structure  $H_L = L \cdot U^1$  on  $H$ , hence a complex conjugation  $\sigma_L$  with fixed points  $LU^1$ .

The disks in  $S$  likewise form a homogeneous space under  $\operatorname{SL}(2, \mathbb{C})$ , the stabilizer of the one associated to  $H_{\mathbb{R}}$  being  $\operatorname{SL}(2, \mathbb{R})$ . The correspondence

$$L \mapsto X(L) := \{D \mid \langle d, \sigma_L(d) \rangle > 1 \text{ for } d \text{ a generator of } D\}$$

is therefore a bijection from the space of the real structures under consideration to the space of disks in  $S$ .

Passing from a disk to the complementary disk corresponds, on real structures, to  $L \mapsto iL$ .

## 10

The calculations of no. 3 admit the following interpretation. Let  $X$  be an open disk of  $S$ , corresponding to a real structure  $L$ , whence real forms  $H_L$  of  $H$  and  $\operatorname{SL}(L)$  of  $\operatorname{SL}(2, \mathbb{C})$ . Then  $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$  has an  $H_L$ -invariant pre-Hilbert structure; its completion  $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})^\wedge$  is the irreducible unitary representation of  $H_L$ . One has

$$\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0}) \subset \Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})^\wedge \subset \Gamma(X, \mathcal{W}(-\frac{1}{2})^{\nabla=0}).$$

The double cover  $\mathrm{SL}(L)^\sim$  of  $\mathrm{SL}(L)$  acts on  $\mathcal{O}(-\frac{1}{2})|_X$  compatibly with its action on  $\mathcal{O}(-1)$ ; hence  $\mathrm{SL}(L)^\sim$  acts on  $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$ . This is the action that normalizes the action of  $H_L$  and induces the natural action of  $\mathrm{SL}(L)$  on  $H_L$ .

The subrepresentation  $\Gamma(X, \mathcal{W}^0(-\frac{1}{2})^{\nabla=0})$  may be interpreted as the space of real analytic vectors for the action of  $\mathrm{SL}(L)^\sim$ .

This description shows that the projective representation of  $\mathrm{SL}(L)$  extends to the monoid of  $g \in \mathrm{SL}(2, \mathbb{C})$  such that  $gX \supset X$ .

## 11

Everything preceding generalizes from  $\mathrm{SL}(2, \mathbb{C})$  to  $\mathrm{Sp}(2n, \mathbb{C})$ : replace  $\mathbb{C}^2$  by  $\mathbb{C}^{2n}$ , endowed with a nondegenerate alternating form  $\langle \cdot, \cdot \rangle$  real on  $\mathbb{R}^{2n}$ , replace  $H$  by  $\mathbb{C}^{2n} \times \mathbb{C}^*$  with group law

$$(x, \lambda)(y, \mu) = (x + y, \lambda\mu \exp(\pi i \langle x, y \rangle)),$$

and replace  $S$  by the space of Lagrangian, i.e. maximal isotropic, subspaces  $D$  of  $\mathbb{C}^{2n}$ . The disk attached to a real structure becomes the Siegel space of those  $D$  for which  $|\langle d, \bar{d} \rangle| > 1$  for  $d \neq 0$  in  $D$ . The invertible sheaf  $\mathcal{O}(-\frac{1}{2})$  must be replaced by  $\det(D)^{1/2}$ .

## 12. Remark

Beside representations of  $H$  in which  $\mathbb{C}^*$  acts by multiplication by  $\lambda$ , consider likewise representations in which the action is by  $\lambda^{-1}$ . One defines, as in no. 2, a holomorphic bundle  $\mathcal{W}_-^0$  of such representations over  $S$ , and on  $\mathcal{W}_-^0(-\frac{1}{2})$  one again has a connection, given by a heat equation.

On a connected simply connected open subset  $U$  of  $S$ , consider  $W = \mathcal{W}^0(-\frac{1}{2})$ , its connection, and the action of  $H$ . In  $\Gamma(U, W)^{\nabla=0}$ , for each  $x \in U$  there is a linear form fixed by the corresponding line, defined up to a factor:  $f \mapsto f(e)$  for  $f \in \mathcal{W}_x^0$ . For each  $x \notin U$ , there exists a vector  $v(x)$  fixed by  $D_x$ ; in the Schrödinger model, and for  $x$  corresponding to  $E =$  the  $x$ -axis, it is the constant 1. It is unique up to a factor.

Fix the real structure  $H_{\mathbb{R}}$ , whence a real structure  $S_{\mathbb{R}} \subset S$  (a circle), and a decomposition of  $S - S_{\mathbb{R}}$  into two disks  $U_1, U_2$ . Then:

- (a) The complex conjugate of  $\Gamma(U_1, W)^{\nabla=0}$  is  $\Gamma(U_2, W_-)^{\nabla=0}$ .
- (b) The Hermitian form on  $\Gamma(U_1, W)^{\nabla=0}$ , invariant under  $H_{\mathbb{R}}$ , is reinterpreted as a bilinear pairing between  $\Gamma(U_1, W)^{\nabla=0}$  and  $\Gamma(U_2, W_-)^{\nabla=0}$ , invariant under  $H$ .

**Question.** Let  $C$  be a Jordan curve in  $S$ , separating  $S - C$  into two open sets  $U_1$  and  $U_2$ . Does there exist a unique separating  $H$ -invariant bilinear form pairing  $\Gamma(U_1, W)^{\nabla=0}$  and  $\Gamma(U_2, W_-)^{\nabla=0}$ , induced by a perfect duality between  $\Gamma(U_1, W)^{\nabla=0}$  and  $\Gamma(U_2, W_-)^{\nabla=0}$ ?

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